

CHAPTER 8:

Currency, Interest Rate,
Swaps

Chapter Outline

- Overview
- Interest rate swaps
- Currency swaps

Overview

- Purpose of derivatives and swaps
- Reference rates

Economic Advantage of Swaps

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Swaps create economic value because:

- Firms have comparative advantages in different currencies and rate types (i.e., fixed v floating rates).
- Swaps allow firms to:
 - Borrow where they can get the cheapest rates;
 - Exchange cash flows to match their needs.
- This leads to:
 - Lower borrowing costs;
 - Efficient FX risk management, or interest rate risk management;
 - Access to long-term foreign currency funding (currency swaps).

Purpose of Derivatives and Swaps

- ➡ Help MNCs manage interest-rate risk, exchange-rate risk, credit risk, and combinations of those risks.
- ➡ As of 2022, there were
 - Interest rate swaps $\approx$ \$405.5 trillion (International Swaps and Derivatives Association),
 - Currency swaps $\approx$ \$97 trillion (Bank for International Settlements), and
 - Students do not need to know the following for their quiz or exam. However, it's essential to recognise this significant instrument in the market, including a crucial role that it played as a warning system during the global financial crisis from 2008 to 2010.
 - Credit default swaps = \$3.9 trillion (International Swaps and Derivatives Association).

A credit swap, or more specifically a credit default swap (CDS), is a financial derivative that functions as insurance against the risk of default on a debt security, where the buyer pays periodic premiums to the seller, who then compensates the buyer if the underlying debt issuer defaults.

Reference Rates

- ➔ Derivatives and swaps rely on a reference interest rate.
- ➔ LIBOR (London Inter-Bank Offered Rate) used to be the most common reference rate.
 - The average interest rate set by global banks in London, calculated once a day at 11 a.m. London time.
 - 35 different rates (7 maturities x 5 currencies)
- ➔ Currently LIBOR being phased out by most countries, but no clear leading alternative, although many alternatives exist.
- ➔ The term *risk-free rate* (RFR) is used to refer to alternative reference rates.
 - For the U.S. dollar, the main RFR is the **Secured Overnight Financing Rate (SOFR)**, which is published by the Federal Reserve Bank of New York and based on actual transactions in the Treasury repurchase market, where banks offer Treasury securities as collateral to raise cash.

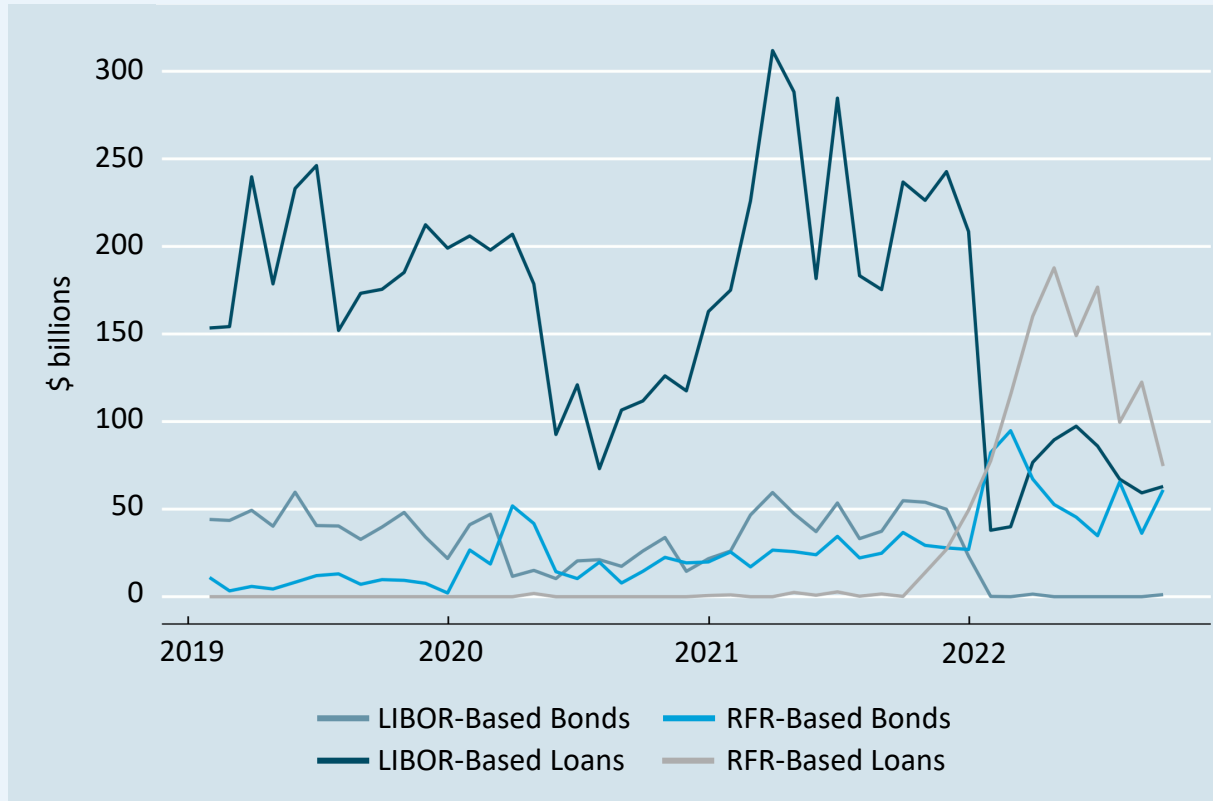
Maturities available:	Overnight 1 week 1, 2, 3, 6, and 12 months
Currencies available:	CHF, EUR, GBP, JPY, USD

Some more information about LIBOR and SOFR

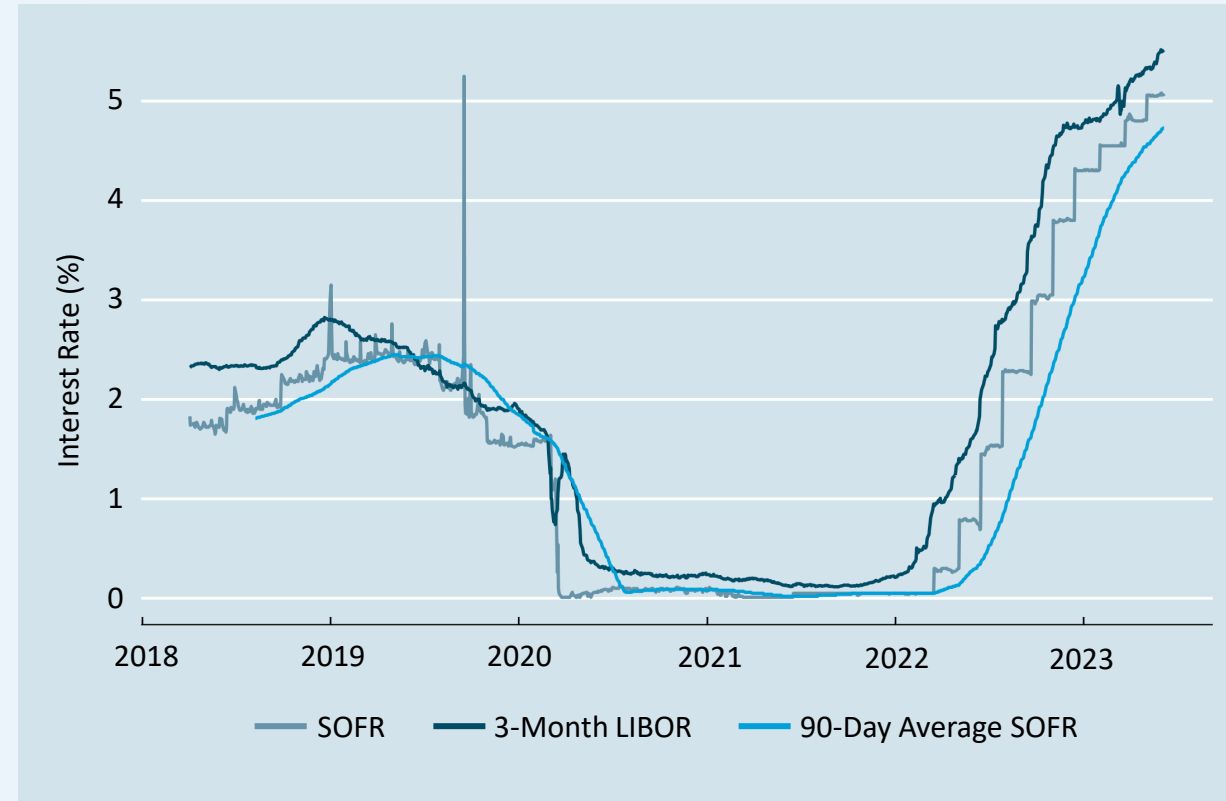
- Please note that LIBOR (London Interbank Offered Rate) is an interest rate average calculated from estimates submitted by the leading banks in London. However, LIBOR is no longer used as a benchmark for the cost of borrowing cash overnight. Instead, SOFR (Secured Overnight Financing Rate) is used.
- SOFR, is a benchmark interest rate used in the US dollar market and is based on actual transactions in the US Treasury repurchase (repo) market.
- SOFR is a broad measure of the cost of borrowing cash overnight, secured by U.S. Treasury securities.
- **How it's calculated:**
- SOFR is calculated from actual transactions in the U.S. Treasury repurchase (repo) market, making it more transparent and reliable than LIBOR, based on bank estimates.
- **Who uses it:**
- SOFR is used as a benchmark for various financial contracts, including loans and derivatives, in the US dollar market.
- **SOFR is administered by:**
- The Federal Reserve Bank of New York
- **SOFR is published:**
- The rate for a given business day is published at 8am Eastern Time on the following business day
- The next diagram shows the same information as the previous slide. However, instead of using SOFR which is now used, it uses the old LIBOR as a floating rate.

Reference Rates

New issuances of bonds and loans using LIBOR and RFRs



SOFR and the 3-Month LIBOR

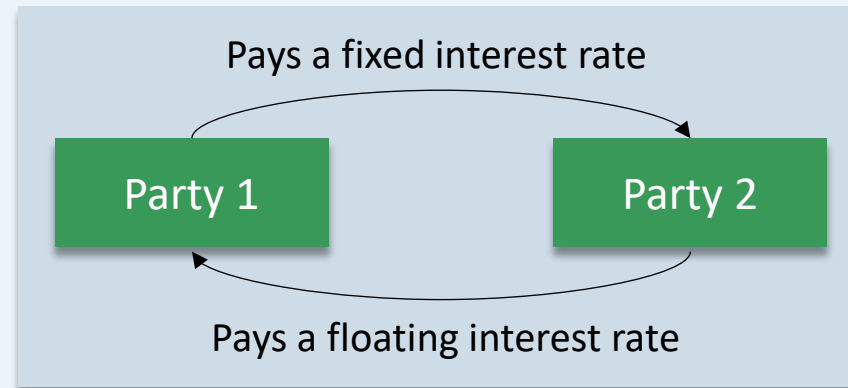


Interest Rate Swaps

- What is an interest rate swap?
- Classic swap transaction
- Interest rate swap: Example

What is an Interest Rate Swap?

- ➔ **Interest rate swap:** an agreement between two parties to exchange the interest payments of the same maturity on a **notional principal** (a reference amount against which the interest rate is calculated)
- ➔ No principal amount changes hands.
 - **Coupon swap:** Party 1 pays a **fixed** interest rate to Party 2, who pays a **floating** interest rate to Party 1.



- The fixed rate is calculated as a percentage of the notional principal and does not change over time.
- The floating rate is periodically reset based on a spread over LIBOR or other variable RFR (e.g., fed funds rate, etc.).

The Classic Interest rate Swap Transaction

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- Counterparties A and B both require \$100 million for a five-year period. To reduce their firms' financing risks, counterparty A would like to borrow at a fixed rate, whereas counterparty B prefers to borrow at a floating rate.
- A is a company with a BBB rating and B is a AAA rated bank. Firm A can raise funds via a bank but it finds it difficult to raise fixed rate funds from bond issues in the capital market at an attractive rate. In contrast, B can borrow at the finest rates in either bond or bank market.

Classic Swap Transaction

Example of a class swap

- ➔ Parties A and B both require \$100 million for a five-year period.
 - Party A (BBB rating) wants to borrow at a fixed rate and qualifies for 8.5%.
 - Party B (AAA rating) wants to borrow at a floating rate and qualifies for a SOFR6.

Borrower	Fixed rate	Floating rate
Counterparty A (BBB)	8.50%	6-month SOFR + 0.5%
Counterparty B (AAA)	7.00%	6-month SOFR
Difference	1.5%	0.5%

100-basis-point differential

- ➔ The parties want to take advantage of the 100-basis-point spread differential by entering into a swap.

Classic Swap Transaction

➔ The parties enter into a swap

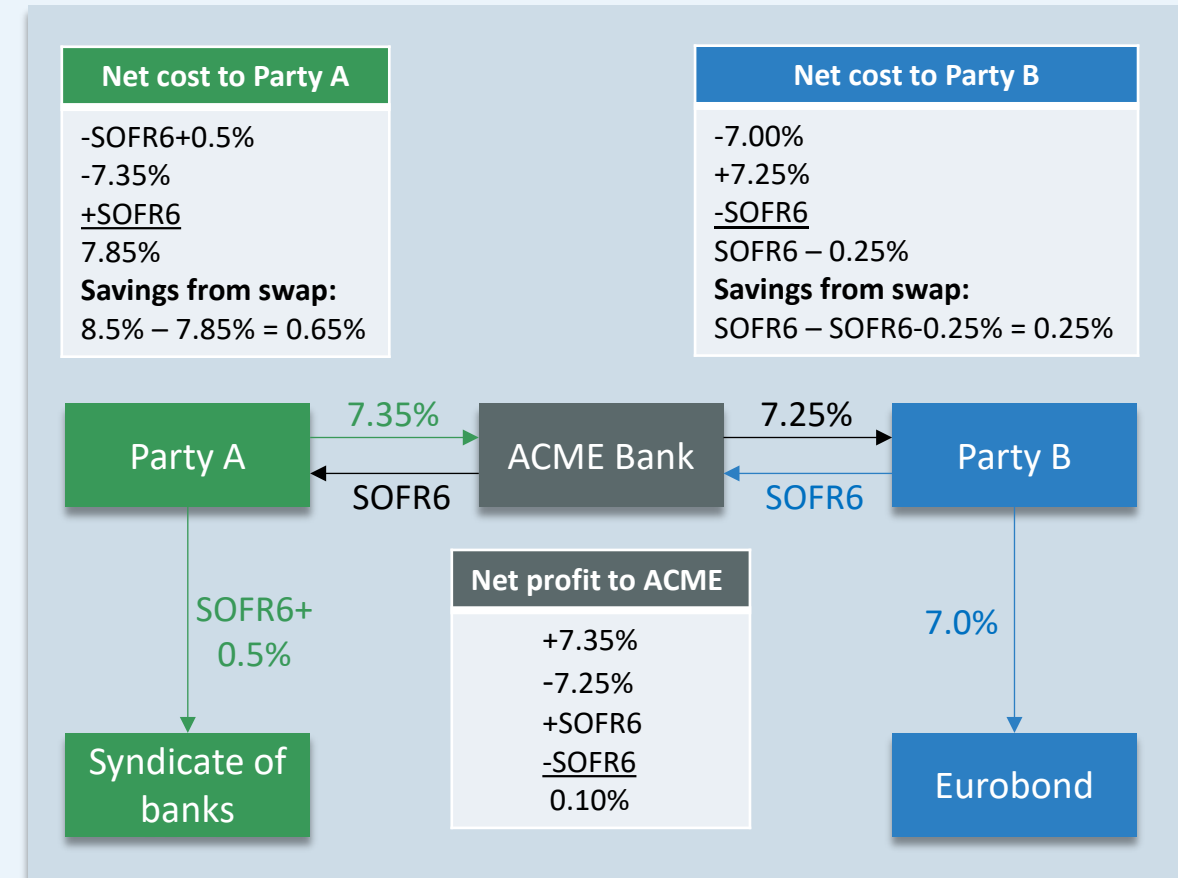
➔ Step 1: The parties borrow at rates contrary to the terms they want:

- ➔ Party A, which wants a fixed rate, takes out a floating-rate Eurodollar loan from a syndicate of banks at $\text{SOFR6} + 0.5\%$.
- ➔ Party B, which wants a floating rate, issues a 7.0% fixed Eurobond.

➔ Step 2: The parties swap their interest payments

- ➔ Party A pays ACME Bank 7.35% for five years and receives SOFR6 from ACME Bank – converting Party A's floating rate into a fixed.
- ➔ Party B pays the SOFR6 to ACME Bank for five years and receives payments of 7.25% from ACME Bank – converting Party B's fixed rate into a floating rate.

5-year, \$100 million loan



The 100 basis point differential (Note: a basis point is 1/100 of a percentage point) is an actual gain to the two firms.

To see this, we compare the cash-flows with and without the swap:

- Total with swap = $7\% + \text{SOFR} + 0.5\% = \text{SOFR} + 7.5\%$
- Total, NO swap = $8.5\% + \text{SOFR} = \text{SOFR} + 8.5\%$.
- The difference is the basis point differential (or *total gain*):
- $(\text{SOFR} + 8.5\%) - (\text{SOFR} + 7.5\%) = 8.5\% - 7.5\% = 1\%$.

Cost Savings Associated with Swaps

Net Cost to Party A

- (SOFR+0.5%) (pays its bank)
- 7.35% (pays ACME)
- +SOFR (receives from ACME)
- 7.85% (net payment)
- **Savings from swap:**
 $8.5\% - 7.85\% = 0.65\%$

Net Cost to Party B

- 7.00% (pays Eurobond)
- +7.25% (receives from ACME)
- SOFR (pays to ACME)
- SOFR – 0.25% (net payment)
- **Savings from swap:**
 $\text{SOFR} - (\text{SOFR} - 0.25\%) = 0.25\%$

Cost Savings Associated with Swaps

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Party	Normal Funding Cost (%)	Cost After Swap (%)	Difference (%)
Counterparty A	8.50	7.85	0.65
Counterparty B	SOFR	SOFR -0.25	0.25
BigBank	—	—	0.10
Total			1.00

Currency Swaps

- What is a currency swap?
- Classic currency swap
- Fixed-for-fixed currency swap
- Fixed-for-floating currency swap

The Currency Swap

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Definition:

Two parties exchange foreign currency-denominated debt at periodic intervals.

It is an exchange of debt-service obligations (coupon payments + principal) denominated in one currency in exchange for coupon payments + principal on an amount of debt denominated in another currency.

The two parties are able to replace cash flows denominated in one currency with cash flows in a more desired currency.

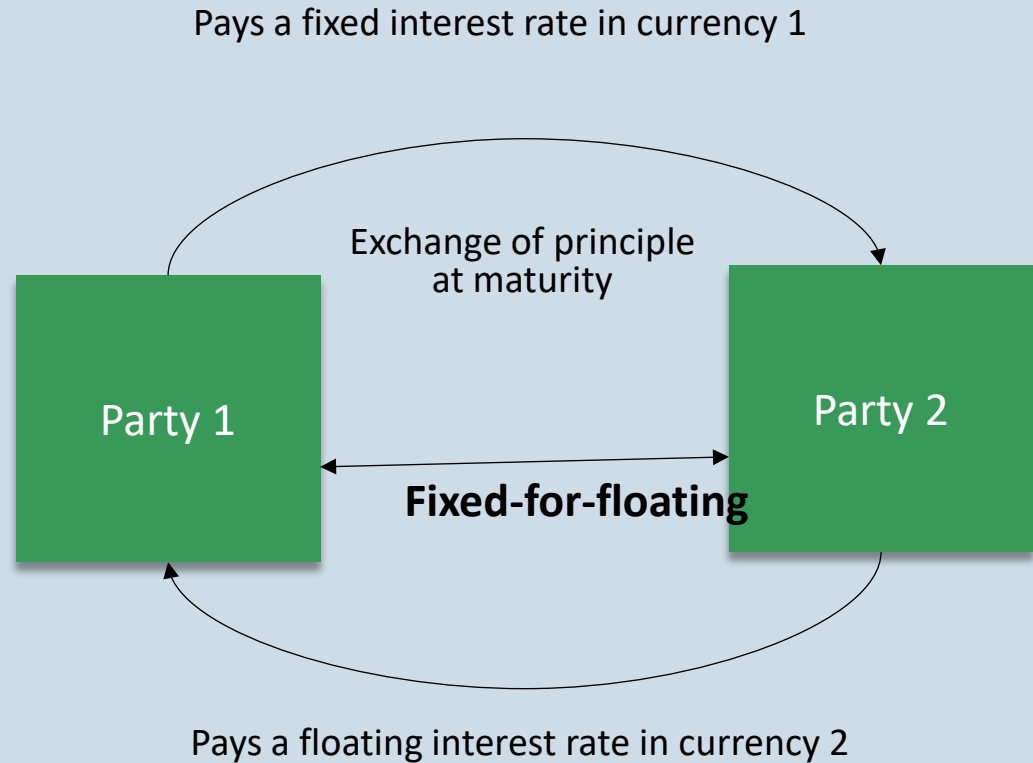
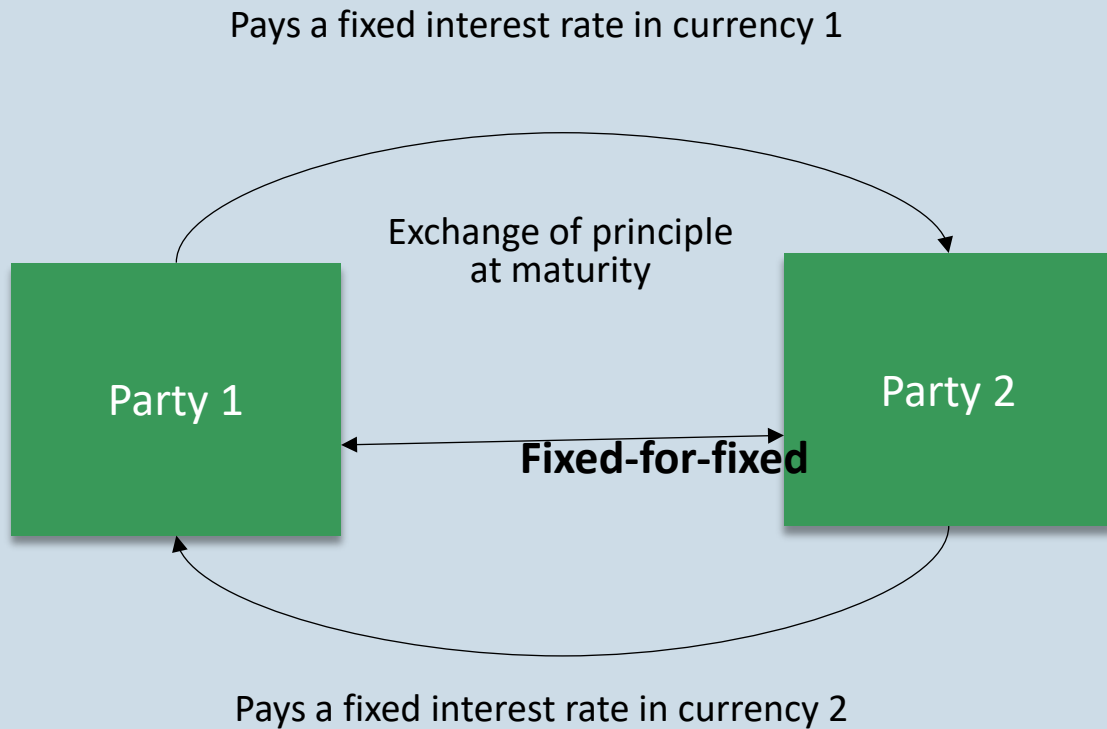
The principal amounts are chosen to be equal at the current exchange rate. They also swap the initial price of the bonds at date 0.

What is a Currency Swap?

- ➔ An exchange between two counterparties of interest payments denominated in different currencies to manage both interest rate and exchange rate risk.
 - **Fixed-for-fixed** currency swap: the interest payments exchanged are both fixed rates, denominated in different currencies. The main purpose is to manage exchange rate risk.
 - **Fixed-for-floating** currency swap: Party A pays Party B a fixed-rate interest payment denominated in one currency while Party B pays Party A a floating-rate interest payment denominated in another currency.
- In contrast to an interest rate swap, there is an exchange of principal amounts at maturity.

3.1 What is a Currency Swap?

- Visual presentation



The Currency Swap

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- Company A has borrowed Japanese yen at a fixed interest rate. It can transform its yen debt into a fully hedged dollar liability by exchanging cash flows with counterparty B.

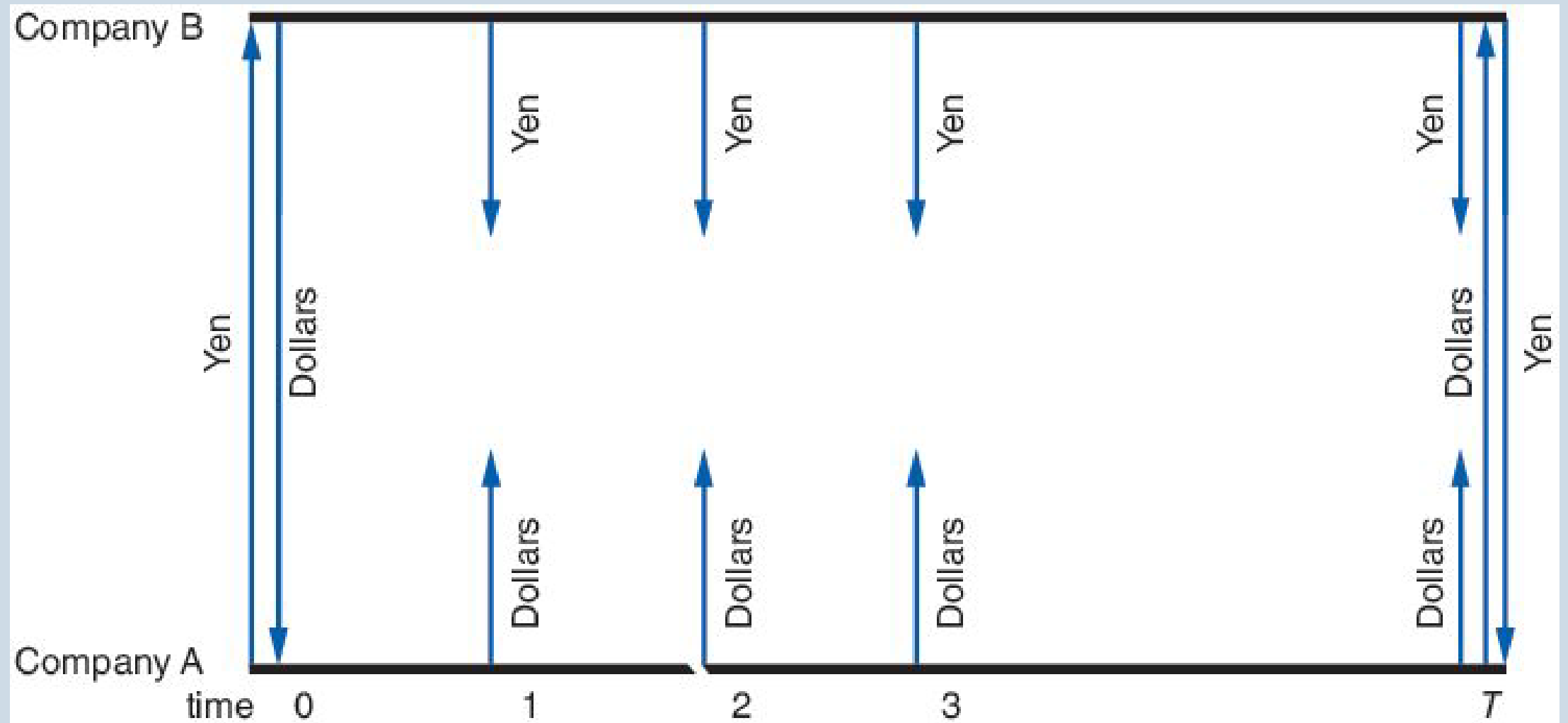
There are parallel interest and principal repayment schedules. At each payment date, company A will pay a fixed interest rate in dollars and receive a fixed rate in yen. The counterparties also exchange principal amounts at the start and the end of the swap arrangement.

Basic Currency Swap

- ➡ By swapping their future cash-flow obligations, the counterparties can replace cash flows denominated in one currency with cash flows in a more desired currency.
- ➡ E.g., if Company A has borrowed Japanese yen at a fixed rate, it can transform its yen debt into a fully hedged dollar liability by exchanging cash flows with Company B.
- ➡ The two loans have parallel interest and principal repayment schedules.
- ➡ At each payment date, Company A pays a fixed rate in dollars and receives a fixed rate in yen.
- ➡ The parties also exchange principal amounts at the start and end of the swap arrangement.
- ➡ Please see the next slide for the visual presentation

Diagram of a fixed-for-fixed currency swap

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DowDupont Swaps fixed for fixed with michelin

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- DowDupont is looking to hedge some of the euro exposure by borrowing in Euros.
- French Tire manufacturing Michelin is seeking dollars to finance additional investment in the US Market.
- Both want the equivalent of \$200 million in fixed-rate financing for 10 years.
- Dow can issue dollar-denominated debt at a coupon rate of %7.5 in dollar or Euro-denominated debt at a coupon rate of 8.25%
- Equivalent rates for Michelin are 7.7% in dollars and 8.1% in Euros.
- Given the fact that they have similar credit ratings, the best way for them to borrow in the other's currency is to issue debt in their own currencies and then swap the proceeds and future debt-service payments

Dow Chemical Swaps fixed for fixed with michelin

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- Assuming a current spot rate of Euro1.1/\$, Michelin would issue €220 million in 8.1% in euros. DowDupont would float a bond issue of \$200 million at 7.5%. (over 10 years)

The coupon payment on these bond issues are €17,820,000 ($.081 \times \text{€}220 \text{ million}$) and \$15,000,000 ($0.075 \times \$200 \text{ million}$), respectively, giving rise to the following debt-service payments

Example of Fixed for Fixed Euro-U.S. Dollar Currency Swap (cont.)

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- The coupon payment on these bond issues are €17,820,000 ($.081 \times \text{€}220$ million) and \$15,000,000 ($0.075 \times \200 million), respectively, giving rise to the following debt-service payments, as shown below

Year	Michelin	Dow Chemical
1-10	€17,820,000	\$15,000,000
10	€220,000,000	\$200,000,000

3.4 Fixed-for-Fixed Currency Swap

➔ Suppose US-based chemical company DowDupont wants to borrow \$200 million in euros and France-based tire maker Michelin wants to borrow \$200 million. Both want a fixed rate and maturity of 10 years.

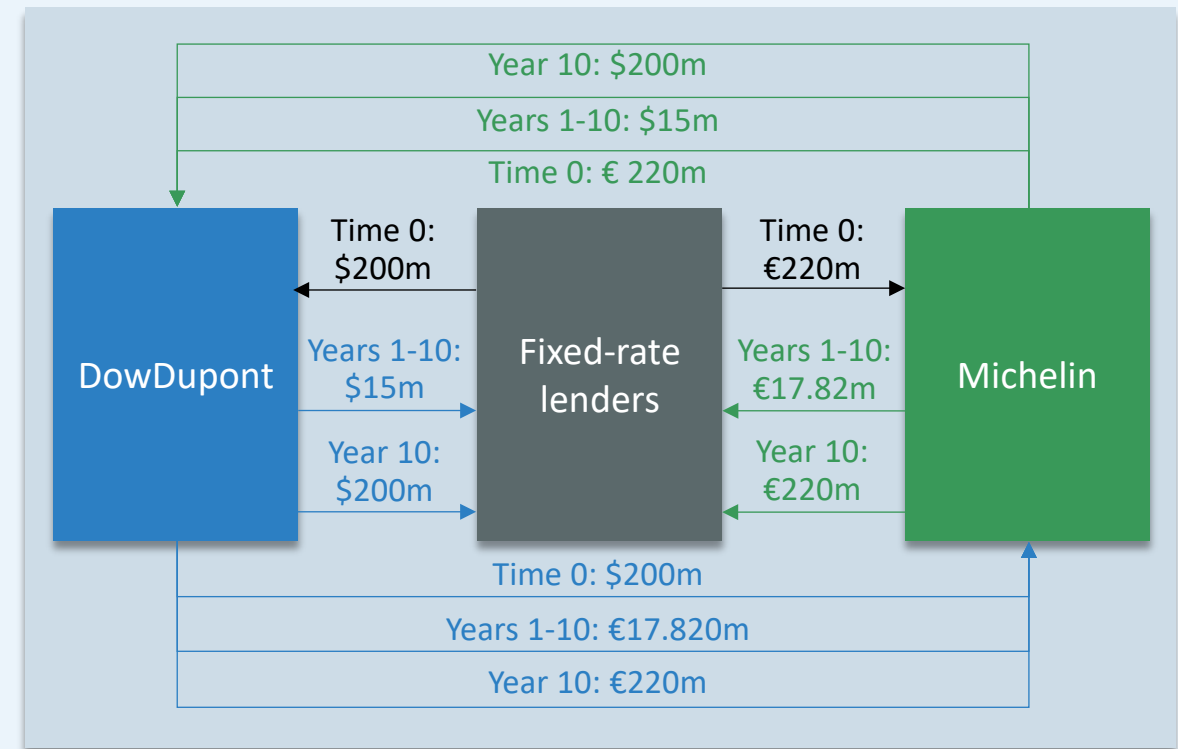
➔ Borrowers can issue debt at the following rates:

Borrower	Dollar fixed rate	Euro fixed rate
Michelin	7.70%	8.10% → Better for DowDupont
DowDupont	7.50%	8.25%
Difference	0.20% ← Better for Michelin	-0.15%

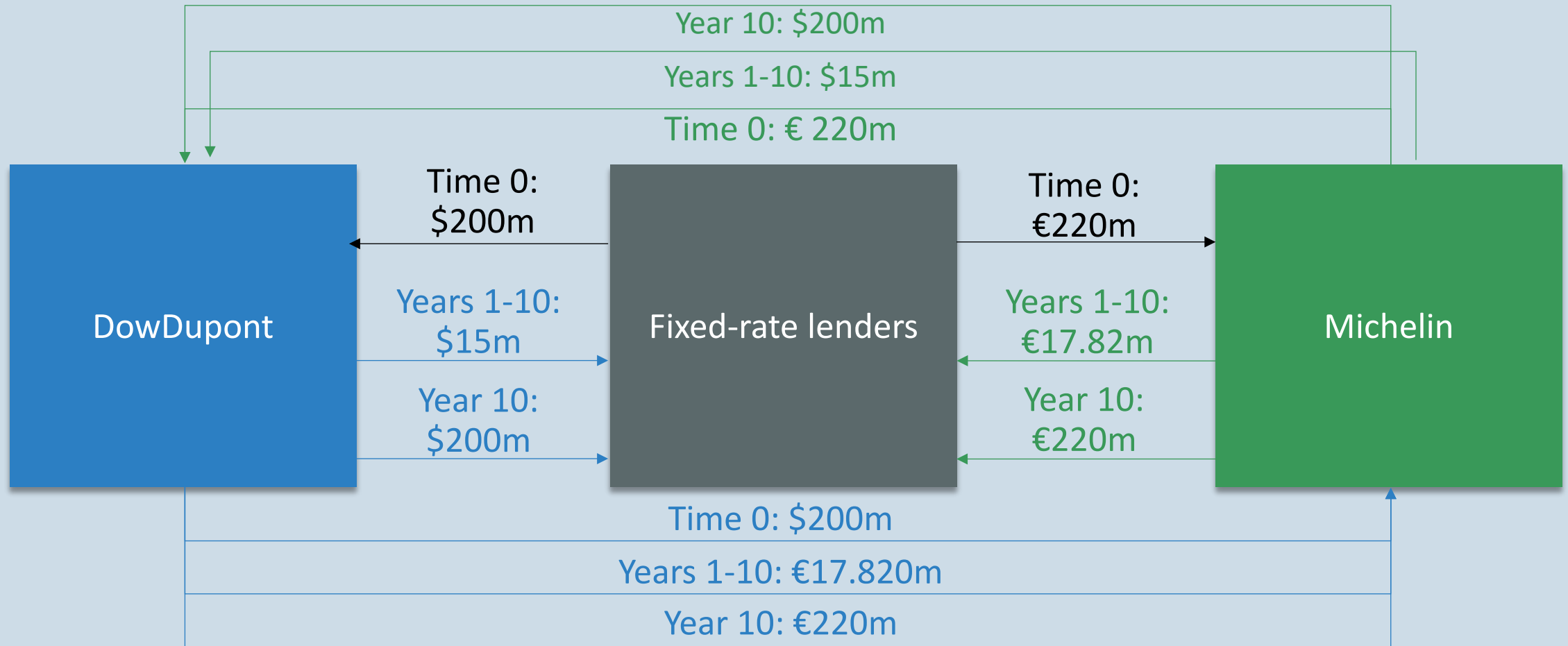
➔ Michelin and DowDupont have comparative advantages at borrowing in euros and dollars, respectively.

➔ They can therefore swap the proceeds and the payment flows to benefit from the basis-point differential.

Fixed-for-fixed payment flows
(spot rate = \$/€)



Fixed-for-fixed payment flows
(spot rate = \$/€)



Swaps fixed for floating with Michelin²⁸

DowDupont prefers to borrow floating rate euro instead of fixed rate euros, whereas Michelin prefers fixed rate dollars. Dow can borrow flexible rate at $\text{SOFR} + 0.35\%$ whereas the cost of borrowing for Michelin is $\text{SOFR} + 0.125\%$.

Fixed rate for Dow is 7.5% and for Michelin 7.7%

3.3 Fixed-for-Floating Currency Swap

Fixed-for-floating payment flows

- ➔ Suppose DowDupont prefers to borrow floating-rate euros instead of fixed-rate euros, whereas Michelin still wants fixed-rate dollars.
- ➔ The best way for them to achieve their currency and interest rate objectives is to issue debt in their own currencies and then swap the proceeds and future debt-service payment.

Borrower	Dollar fixed rate	Euro floating rate
Michelin	7.70%	SOFR + 0.125%
DowDupont	7.50%	SOFR + 0.35%
Difference	0.20%	-0.15%

